

Raw Sockets and Packet Layouts in Unicon

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Abstract

Unicon already opened TCP and UDP sockets through `open()`, but IP-level protocols such as ICMP and IGMP were not supported. This report describes raw sockets: mode "nr" opens `SOCK_RAW`, `proto=` selects the IP protocol number, and `writes()/receive()` carry upper-layer bytes on the same datagram path UDP already uses. The runtime does not assemble protocol payloads. Package `net` supplies `PacketSpec`, a declarative engine for fixed binary layouts, plus ICMP echo and IGMPv2/v3 classes and session wrappers.

Keywords: Unicon, raw sockets, `SOCK_RAW`, ICMP, IGMP, `PacketSpec`, runtime, technical report.

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1 1. Introduction

This report describes raw IP sockets in Unicon: how a handle is opened, how datagrams are sent and received, how protocol bytes are built and decoded, and what remains out of scope. It is formatted as a Unicon Technical Report [1]. Socket attributes (`ttl`, `iface`, `join`, and the rest) are the multicast facility [2]; this report adds mode `nr` and the packet library on top of that. The language-facing reference lives in *Programming with Unicon* [3] (Chapter 5, "Raw Sockets" and "PacketSpec and package net"; Chapter 14, `uping` and `uigmp`) and the language reference (`open` mode `nr`, attributes `proto` and `hdrincl`, subscript `peek`). Automated plumbing and peek tests live in `tests/posix/raw_sock.icn`.

Raw sockets are part of the POSIX network facility. A build without POSIX networking has no `n` modes at all. Opening `SOCK_RAW` usually needs elevated privileges (root or administrator); that is an operating-system restriction, not a Unicon one.

2 2. Motivation

TCP and UDP cover most application protocols. Diagnostic and control traffic sits one layer down: ICMP echo [4], IGMP membership [5] [6], OSPF, GRE, PIM. Those messages are IP protocol payloads, not port-addressed transport streams. ICMP, IGMP, and other raw IP protocols were not supported.

The rest of Unicon networking already treats `open()` as the constructor and `Attrib()` as the option verb. Raw sockets follow that pattern instead of adding `rawsocket()` or a parallel packet API. Payload construction is a library problem: the runtime delivers bytes, and `PacketSpec` names the fields.

3 3. Design principles

r after n is the socket type, like u for UDP. Mode letter `r` remains ordinary read mode unless a preceding `n` has already marked the handle as a network socket. Then "`nr`" means `SOCK_RAW`, the same way "`nu`" means `SOCK_DGRAM`. "`rn`" is not a raw socket: `r` is consumed as read mode before `n` is seen.

`proto=` is required and is fixed at `socket()` time. The third argument to `socket(2)` is the IP protocol number. There is no useful default: ICMP, IGMP, and OSPF are different sockets. A missing, empty, or unknown `proto=` is error 1310 (bad socket attribute). `f["proto"]` peeks the protocol number stored at `open()`; `Attrib()` cannot change it.

The runtime does not build protocol payloads. Writes send whatever bytes the caller supplies. Receives return whatever the kernel delivered, often including an IPv4 header on Linux. `import net` and `PacketSpec` (or hand-assembled strings) fill that gap.

Datagram I/O matches UDP. Raw sockets are not connected. `writes()` uses `sendto(2)` with the destination saved at `open()`. Incoming packets use `receive()`, which returns a record with `addr`, `msg`, `saddr`, `daddr`, and (for IPv4 raw datagrams that include a header) `ttl` and `proto`. Those extra fields are also available as `r.ttl` / `r.saddr`. Use `receive()`, not `read()` / `reads()`.

Two layers for callers. Session wrappers (`Ping`, `Igmp`) hide mode letters for the common cases. `PacketSpec` classes are there when the program must craft or classify bytes itself.

4 4. Opening a raw socket

```
f := open("8.8.8.8", "nr", "proto=icmp", "ttl=64") |
    stop(&errortext)
```

The first argument is a bare host or address, not `host:port`. Raw destinations are hosts. A single `host:port` form is accepted when there is exactly one colon and a non-empty port, but ICMP and IGMP do not use ports. IPv6 literals contain colons; they are treated as hosts and are not split as `host:port`.

`proto=` takes a symbolic name or an integer 0–255:

Name	Protocol
icmp	IPPROTO_ICMP

Name	Protocol
icmpv6 / icmp6	IPPROTO_ICMPV6 (fails if the host has no constant)
igmp	IPPROTO_IGMP
tcp / udp	IPPROTO_TCP / IPPROTO_UDP
gre	47
ospf	89
pim	IPPROTO_PIM
raw	IPPROTO_RAW
0 .. 255	that protocol number

Names are case-insensitive. Unknown names and numbers outside 0–255 fail at `open()` with 1310.

If the kernel refuses `SOCK_RAW`, `open()` fails and `&errortext` is the system message (typically "Operation not permitted" or "Permission denied"). That is success of the Unicon plumbing and a privilege problem on the host.

```

procedure main()
  local f
  if f := open("8.8.8.8", "nr") then
    write("oops missing proto")
  else
    write("missing-proto: ", &errortext)
    if f := open("8.8.8.8", "nr", "proto=nope") then
      write("oops bad proto")
    else
      write("bad-proto: ", &errortext)
      if f := open("8.8.8.8", "nr", "proto=icmp") then
        write("raw-icmp: ok")
      else
        write("raw-icmp: ", &errortext)
    end
  end
end

```

Output:

```

missing-proto: bad socket attribute
bad-proto: bad socket attribute
raw-icmp: Operation not permitted

```

The last line is the unprivileged case. With privileges it prints `raw-icmp: ok` (or the test suite's `plumbing-ok`).

5 5. Attributes and status peek

Trailing `open()` arguments are the same `name=value` socket attributes used for TCP and UDP [2]. Booleans are exactly `yes` or `no`. `ttl` and `iface` were named for this sharing: hop limit and interface are not multicast-only knobs.

Attribute	Role on a raw socket
<code>proto=</code>	Required. Consumed at <code>socket()</code> . Peek with
<code>hdrincl=yes</code>	Sets <code>IP_HDRINCL</code> so writes may include a complete IPv4
<code>ttl=</code>	Unicast and multicast hop limits (<code>IP_TTL</code> /
<code>rcvbuf=</code> / <code>sndbuf=</code>	Socket buffer sizes in bytes.
<code>broadcast=</code> / <code>mcastloop=</code>	Same meaning as UDP.
<code>join=</code> / <code>leave=</code> / <code>source=</code> / <code>iface=</code>	Multicast membership, including SSM <code>source@group</code>

`Attrib(f, "ttl=4")` sets hop limits after `open()`. Status uses the same `[ ] / key()` get as UDP [2]: unknown names raise 1310, unpopulated fields fail. Boolean peek fields that answered succeed with `"yes"` or `&null` (never `"no"`); that is output only – `open()` and `Attrib()` still take `yes` or `no`. Peeking a closed handle is error 174. `key(f)` that has not yet produced a name also raises 174 if the handle is already closed; after the first suspend, `close()` makes the generator fail instead of raising. Raw sockets add `proto` (fixed at `socket()`) and `hdrincl`.

Name	Peek value
<code>proto</code>	IP protocol number from <code>proto=:</code> 1 (icmp),
<code>hdrincl</code>	"yes" or <code>&null</code> (not "no")
<code>ttl</code>	Hop limit (integer), e.g. 64
<code>iface</code>	Outbound / join interface, e.g. 127.0.0.1
<code>groups</code>	List of joined groups, e.g. 239.1. 1.1; unpopulated
<code>mcastloop / reuseaddr / reuseport /</code>	"yes" or <code>&null</code> (not "no")
<code>rcvbuf / sndbuf</code>	Buffer sizes in bytes (integers)

```
f := open("127.0.0.1", "nr", "proto=icmp") | stop(&errortext)
write("proto=", f["proto"])
write("ttl=", f["ttl"] | "unpopulated")
every k := key(f) do
  write("field ", k)
close(f)
```

6 6. Send and receive

```
writes(f, icmp_bytes)
r := receive(f)
write(r.addr, " ", *r.msg, " bytes")
```

`writes()` calls `sock_write()`, which for `SOCK_RAW` (and UDP) does `sendto` to the `addrinfo` saved at `open()`. There is no `connect(2)`. Changing the destination means opening another handle (or, with `hdrincl=yes`, putting the destination in a supplied IP header).

`receive()` peeks with `MSG_PEEK` into a 2K buffer, then consumes with `recvfrom`. If the peek fills 2K, the consume uses a 64K buffer so a large datagram is not truncated. The result is a `posix_message`: `addr` is the peer (often `a.b.c.d:0` for raw IPv4), `msg` is the byte string. `saddr` / `daddr` use the usual IP-header names: on an IPv4 raw receive they are the header source and destination; on UDP, `saddr` is the numeric peer address and `daddr` is unpopulated. When the receive buffer starts with an IPv4 header, `ttl` and `proto` are that header's hop limit and protocol; otherwise those fields are null.

On many IPv4 stacks a raw receive buffer starts with the IP header, then the protocol message. Linux ICMP is the usual example. Strip it before decoding ICMP or IGMP ([section 7](#)). IPv6 raw sockets typically deliver the upper layer only; there is no IPv6 header-strip helper yet.

`select()` works on the handle. Session wrappers use it to implement timeouts around `receive()`.

As with UDP, `read()` / `reads()` are the wrong verbs for datagrams. Use `receive()`.

7 7. PacketSpec

Module `packetspec` (package `net`, `uni/lib/packetspec.icn`) is a small declarative engine for fixed binary layouts. Protocol classes inherit `PacketSpec` and override `define()`.

Layer-1 helpers pack and unpack network-order integers (`u8` / `u16` / `u32` / `u64` and the matching `get_u*` readers) and IPv4 addresses (`ipv4` / `to_ipv4` / `get_ipv4`). `inet_cksum()` is RFC 1071 [7].

7.1 7.1 Declaring a layout

```
class IcmpEchoRequest : IcmpPacket()
  method define()
    field("type", 1, 8)
    field("code", 1, 0)
    checksum_field("checksum")
    field("id", 2)
```

```

        field("seq", 2)
    end
end

```

`field(name, size, defval)` is a network-order integer. `bytes_field` is opaque fixed-width bytes. `ipv4_field` accepts a dotted-quad, integer, or four raw bytes on build and returns a dotted-quad on decode. `checksum_field` is a 16-bit placeholder filled at `build()` time over the header plus payload. Pass `exclude_off` / `exclude_len` (1-based) to omit a range from the sum — OSPF Authentication and PIM Register are the documented cases; those layouts are not in the library yet.

`require(name)` marks a field that `build()` must see via `set()` or the optional extra table. `relax()` / `build(..., lax)` skip that check for incomplete test packets. `strict()` and `clear()` turn checking back on.

7.2 7.2 Build, decode, describe

Preferred call shape (no string-keyed tables required). This needs no raw socket:

```

import net

procedure main()
    local pkt, v
    write(IcmpEchoRequest().describe())
    write(IcmpEchoRequest().set("id", 1).set("seq", 2).describe())
    pkt := IcmpEchoRequest().set("id", 1).set("seq", 2).build(, "hi")
    write("len=", *pkt)
    v := IcmpEchoRequest().decode(pkt)
    write("type=", v["type"], " id=", v["id"], " seq=", v["seq"])
end

```

Output:

```

type:int=8  code:int=0  checksum:checksum(auto)  id:int=0  seq:int=0
type:int=8  code:int=0  checksum:checksum(auto)  id:int=1  seq:int=2
len=10
type=8 id=1 seq=2

```

`set()` and `set_payload()` are chainable. `build(extra, payload, lax)` overlays an optional table, appends payload, and returns the complete byte string with checksum filled. `decode(s)` returns a table of field names to values. `describe()` is a one-line summary for the REPL: names, kinds, defaults, required markers, and current `set()` values.

7.3 IPv4 receive helpers

`strip_ipv4(pkt)` uses the header IHL to return the upper-layer message. `ipv4_ttl(pkt)` and `ipv4_src(pkt)` read TTL and source from the same leading header. ICMP and IGMP classes call `strip_ipv4` before matching.

8. ICMP echo

`uni/lib/icmp.icn` adds echo request (type 8) and echo reply (type 0) [4] on top of `PacketSpec`:

```

PacketSpec
  IcmpPacket      strip_ip / match
    IcmpEchoRequest  type 8; echo(id, seq, payload)
    IcmpEchoReply    type 0; match_echo(pkt, id, seq)
Ping              session wrapper

```

`Ping(host)` opens `proto=icmp` with TTL 64 and a 2000 ms default wait. `echo()` sends a request (56 zero bytes unless a payload is given), waits with `select()`, and succeeds with a decode table plus `rtt` (milliseconds,

real), rtt_us, ttl (from the IPv4 header), and addr. Round-trip time uses `gettimeofday()`.

```
import net

procedure main()
  local p, v
  p := Ping("8.8.8.8") | stop(&errortext)
  write(p.describe())
  if v := p.echo() then
    write("rtt=", v["rtt"], " from ", v["addr"])
  p.close()
end
```

`set_timeout(ms)` changes the default wait. An explicit sequence number and payload can be passed to `echo()`; otherwise the session increments `seq`. The same exchange with an explicit socket:

```
import net

procedure main()
  local f, req, reply, id, pkt, r
  req := IcmpEchoRequest()
  reply := IcmpEchoReply()
  id := iand(?100000 | 1, 16rFFFF)
  f := open("8.8.8.8", "nr", "proto=icmp", "ttl=64") |
    stop(&errortext)
  pkt := req.set("id", id).set("seq", 1).build(, "unicon")
  writes(f, pkt)
  r := receive(f)
  if reply.match_echo(r.msg, id, 1) then
    write("reply from ", r.addr)
  close(f)
end
```

Other ICMP types (destination unreachable, time exceeded, and so on) are not in the library. `proto=icmpv6` opens an ICMPv6 socket, but there are no Neighbor Discovery or echo layouts yet.

9 9. IGMP

`uni/lib/igmp.icn` covers IGMPv2 (RFC 2236) and IGMPv3 (RFC 3376):

```
PacketSpec
  IgmpPacket
    Igmpv2Query / Igmpv2Report / Igmpv2Leave
    Igmpv3Query / Igmpv3Report
  Igmp                                session wrapper
```

IGMPv2 messages are an eight-octet header. Report and leave `require()` a group. IGMPv3 queries add flags, QQIC, and a source list. IGMPv3 reports carry group records: ASM join/leave and SSM source filters (`join_sources`, `allow`, `block`).

`Igmp()` opens `proto=igmp` bound at `0.0.0.0`. `recv(ms)` waits, then returns a table with `kind`, `addr`, `msg`, `len`, and decoded fields when recognized. `kind` is one of `v3_report`, `v3_query`, `v2_query`, `v2_report`, `v2_leave`, or `unknown`.

PacketSpec classes work without a raw socket:

```
import net

procedure main()
  local pkt, v
  write(Igmpv2Report().describe())
  write(Igmpv3Report().describe())
  pkt := Igmpv2Report().report("239.1.1.1")
  write("v2 report len=", *pkt)
  v := Igmpv2Report().decode(pkt)
```

```

    write("type=", v["type"], " group=", v["group"])
    pkt := Igmpv3Report().join("239.1.1.1")
    write("v3 join len=", *pkt)
end

```

Output:

```

type:int=22 max_resp_time:int=0 checksum:checksum(auto)
↪ group:ipv4(required)
type:int=34 reserved1:int=0 checksum:checksum(auto)
↪ reserved2:int=0 nrecords:int=0
v2 report len=8
type=22 group=239.1.1.1
v3 join len=16

```

Opening a live IGMP socket needs the same privileges as ICMP:

```

import net

procedure main()
    local g, v
    g := Igmp() | stop(&errortext)
    write(g.describe())
    if v := g.recv(1000) then
        write(v["kind"], " from ", v["addr"])
        g.report("239.1.1.1")    # IGMPv2 membership report
        g.join("239.1.1.1")     # IGMPv3 ASM join
        g.close()
    end
end

```

Session send helpers: **report** / **leave** / **query** (v2), **join** / **leave3** (v3 ASM), and **send(bytes)** for a hand-built packet. SSM source-filter joins use the `PacketSpec` class, not the session wrapper:

```
pkt := Igmpv3Report().join_sources("239.1.1.1", ["192.0.2.1"])
g.send(pkt)
```

Exported constants (IGMP_MEMBERSHIP_QUERY, IGMP_V3_MEMBERSHIP_REPORT, IGMP_CHANGE_TO_EXCLUDE_MODE, and the rest) are initialized on first IgmpPacket construction.

10 10. Sample programs

uni/progs/uping and uni/progs/uigmp are teaching clients, not replacements for ping(8) or tcpdump(1). They need SOCK_RAW privileges:

```
unicon uping
unicon uigmp
sudo ./uping host [count]
sudo ./uigmp [seconds]
sudo ./uping -v host      # also show open()/Ping API demos
sudo ./uigmp -v           # also show open()/Igmp API demos
```

Default mode uses the session wrappers. -v / -d also runs the lower-level open(..., "nr", ...) plus PacketSpec path and prints describe() output.

```
sudo ./uping 192.0.2.1
```

Output:

```
PING 192.0.2.1 (192.0.2.1): 56 data bytes
64 bytes from 192.0.2.1: icmp_seq=0 ttl=118 time=18.390 ms
64 bytes from 192.0.2.1: icmp_seq=1 ttl=118 time=16.245 ms
64 bytes from 192.0.2.1: icmp_seq=2 ttl=118 time=15.533 ms
64 bytes from 192.0.2.1: icmp_seq=3 ttl=118 time=17.311 ms
--- 192.0.2.1 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max = 15.533/16.869/18.390 ms
```

```
sudo ./uigmp
```

Output:

```
listening on IGMP

36 bytes from 192.0.2.10: IGMPv3 query group=0.0.0.0 max_resp=100
↳ qrv=2
36 bytes from 192.0.2.10: IGMPv3 query group=239.1.1.1 max_resp=100
↳ qrv=2
36 bytes from 192.0.2.20: IGMPv3 report {239.1.1.1
↳ CHANGE_TO_EXCLUDE}
```

11 11. Runtime implementation

The work is in the existing POSIX socket code, not a new file type.

Mode letter. `src/runtime/fsys.r`: after `n` has set `Fs_Socket`, `r` / `R` sets `sock_type = SOCK_T_RAW` and does not set the ordinary read bit. Without `Fs_Socket`, `r` remains `Fs_Read`. `e` (crypto raw-key material) is a later check on the same letter; "`nr`" is a socket, "`er`" is a crypto handle.

Create. `src/runtime/rposix.r: sock_connect` (and the `listen/bind` path) require `proto=` before `socket()`. `uni_getaddrinfo` allows a missing port for `SOCK_T_RAW`. The real protocol number is passed as the third argument to `socket()`; `getaddrinfo` is told protocol 0 so it does not override it.

I/O. UDP and raw share `saddrs[]`: the `addrinfo` from `open()` is kept until `sock_close()`. `sock_write` sends to there. `sock_recv` accepts `SOCK_DGRAM` and `SOCK_RAW`. `proto=` is skipped in `setsockopt` (already consumed). `hdrincl=` sets `IP_HDRINCL`.

Errors. Unknown or missing socket attributes use error 1310 with the offending string (`proto`, `proto=nope`, `hdrincl=maybe`). Kernel failures use the system `&errortext`.

There is no `&features` flag named "raw sockets". If POSIX sockets exist, mode `nr` exists.

12 12. Status and remaining work

The language surface described here is implemented: "`nr`", required `proto=`, `hdrincl=`, `datagram` `writes / receive`, `PacketSpec`, ICMP echo, IGMPv2/v3, `uping`, and `uigmp`.

More PacketSpec layouts. GRE, OSPF, and PIM names are accepted at `socket()` and the checksum helper documents their exclude ranges, but there are no protocol classes yet.

ICMPv6. `proto=icmpv6` opens the socket. Echo, Neighbor Discovery, and an IPv6 header strip are not in `net`.

IPv6 receive metadata. `strip_ipv4 / ipv4_ttl` assume a leading IPv4 header. IPv6 raw receives need their own helpers.

read() on raw handles. UDP clears the read bit so `read()` fails. Raw currently keeps `Fs_Read` because it shares the TCP-like status assignment. The documented API is still `receive()`. Aligning the status bits with UDP would make misuse fail earlier.

Privileged tests. `tests/posix/raw_sock.icn` checks plumbing without sending packets. A live ICMP echo test would need root in CI and a reachable responder; it is not in the suite.

Windows. Opening `SOCK_RAW` is possible with administrator rights, but Windows historically restricts which protocols a raw socket may use. Treat `uping / uigmp` as UNIX-first tools.

Header inclusion. `hdrincl=yes` is wired. There is no `PacketSpec IPv4` header class to go with it; callers who craft full datagrams build those bytes themselves.

13 Appendix: test coverage

`tests/posix/raw_sock.icn` does not require `SOCK_RAW` success. It checks that:

- missing, empty, and unknown `proto=` fail with 1310
- `proto=icmp`, `gre`, `ospf`, and numeric 89 reach `socket()` (success or a privilege denial both print `plumbing-ok`)
- `hdrincl=maybe` is rejected
- without a preceding `n`, mode `"r"` still opens a file for reading
- when `SOCK_RAW` is allowed, `f["proto"]` is 1 after `proto=icmp` and `key(f)` yields populated names (privilege denial still prints `raw-peek:ok` / `raw-key:ok` so the `.std` stays portable)

Expected output is `tests/posix/stand/raw_sock.std`:

```
missing-proto: bad socket attribute
bad-proto: bad socket attribute
empty-proto: bad socket attribute
raw-icmp: plumbing-ok
raw-gre: plumbing-ok
raw-ospf: plumbing-ok
raw-numeric: plumbing-ok
bad-hdrincl: bad socket attribute
read-mode-r: ok
raw-peek:ok
raw-key:ok
```

On an unprivileged account the `raw-*` lines still print `plumbing-ok`: the test treats both a successful `socket()` and a privilege denial as passing plumbing. The `posix` suite picks up every `*.icn` in that directory.

14 References

References

- [1] Clinton Jeffery. "How to Write a Unicon Technical Report". doc/utr/utr15.tex. 2013.
- [2] Jafar Al-Gharaibeh. "Multicast and Socket Attributes in Unicon". doc/utr/utr28.rst. 2026.
- [3] Clinton Jeffery and Shamim Mohamed and Jafar Al-Gharaibeh. "Programming with Unicon". doc/book/. 2026.
- [4] J. Postel. "Internet Control Message Protocol". RFC 792. 1981.
- [5] W. Fenner. "Internet Group Management Protocol, Version 2". RFC 2236. 1997.
- [6] B. Cain and S. Deering and I. Kouvelas and B. Fenner and A. Thyagarajan. "Internet Group Management Protocol, Version 3". RFC 3376. 2002.
- [7] R. Braden and D. Borman and C. Partridge. "Computing the Internet Checksum". RFC 1071. 1988.